

Final ChatGPT Neuroscience Audit

Cognitive Audit Report

Evaluating ChatGPT-4o Through a Neuroscientific Lens

ITAI 4374: Neuroscience as a Model for AI | Spring 2026
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System Profile

ChatGPT-4o is a multimodal large language model developed by OpenAI, accessed through the standard chat interface across multiple isolated sessions. It generates responses by predicting the most contextually appropriate continuation of an input sequence, processing every token simultaneously through self governing mechanisms that compute relational weights across the full input. The model has no sensory system, no body, and no motor system to test predictions against physical experience. It doesn't pass time between sessions. Each new conversation begins as a complete blank. These are architectural facts that matter enormously when comparing a language model against a brain that evolved to navigate a physical world in continuous time. The audit treated these facts as the baseline and built every experiment around them.

Overview

This report is a cognitive audit of ChatGPT-4o conducted across five modules of ITAI 4374: Neuroscience as a Model for AI. It expands directly on the midterm audit submitted earlier this semester and represents the full record of that investigation. The premise driving this audit is that large language models didn't emerge from a vacuum. They were built by researchers who studied the brain and borrowed from it. Sometimes deliberately and sometimes without fully realizing how deep the parallel ran. This report takes that premise seriously by treating

ChatGPT-4o not as a chatbot but as a cognitive system worthy of rigorous evaluation against the biological systems this course described. The audit covers memory and learning, attention and consciousness, decision making, world models and goal-directed behavior, and neurosymbolic reasoning. Each module produced a distinct set of experiments designed to probe a specific brain function and compare observed behavior against the corresponding biological mechanism. What follows is the complete record of that investigation.

Experiment Results

Module 06 — Memory and Learning

Memory and learning were the first systems examined in the midterm audit and returning to them now, after working through decision making, world modeling, and neurosymbolic reasoning, changes what feels significant about the original findings. The midterm established the basic architecture void clearly. ChatGPT-4o has a working memory analog and a semantic memory analog. It has no episodic memory. That conclusion still holds. What the midterm couldn't fully appreciate is how that absence compounds across every other cognitive function the later modules examined. At the time, the cross-session continuity failure felt like a contained limitation. A note about memory, filed and moved past. After Module 09 it reads differently. The hippocampus is more than a memory system. It's a scene construction engine that uses stored experience to simulate the future (McManus, 2026d). Every world model the brain builds is built from consolidated episodic material. ChatGPT-4o can simulate a counterfactual 1600 day and age with impressive coherence within a session. It cannot remember that simulation the next day. The working memory findings from the midterm remain the model's clearest cognitive strength. Within a session, ChatGPT-4o holds and applies context with consistency that mirrors prefrontal function accurately. Semantic generalization also held up across repeated testing. What the

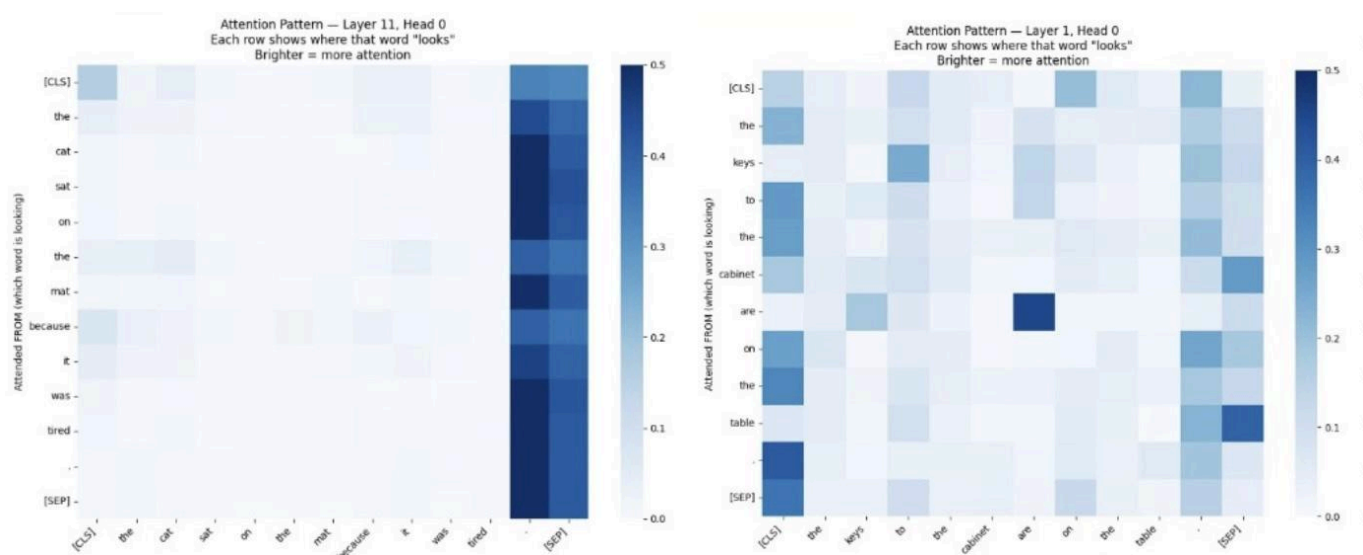
midterm described as a contained positive finding now looks like the ceiling of what a system without hippocampal consolidation can achieve. Strong within the moment. No thread connecting one moment to the next.

Module 07 — Attention and Consciousness

The attention experiments from the midterm produced some of the cleanest results of the entire audit and the passage of time hasn't complicated them so much as sharpened what they reveal. The midterm found that ChatGPT-4o excels at executive attention and fails entirely at passive background monitoring. The Buried Instruction experiment confirmed top-down goal-directed attention. The Prompt Injection experiment confirmed resistance to distraction. The Cocktail Party experiment confirmed the absence of any biological alerting mechanism. The BERT visualization connected transformer depth to the cortical processing hierarchy. What the midterm conceptualized as a set of isolated findings now reads as a single coherent portrait. ChatGPT-4o is a system of deliberate, voluntary, top-down attention with no passive layer underneath it. The brain runs both simultaneously. The executive network handles goal-directed tasks while the alerting network continuously scans the environment for signals worth interrupting for (McManus, 2026b). One operates in the foreground and one operates in the background and they communicate constantly. ChatGPT-4o has only the foreground. The Cocktail Party experiment showed this clearly when a sentence mentioning Jordan from TechCorp by name passed through the model without triggering any response. There was no background monitor to catch it because nothing in the architecture was running beneath the primary task.

The reason this matters more now than it did at midterm is that the later modules revealed how often biological cognition depends on that background layer. Somatic markers in Module 08 operate below conscious reasoning. Predictive error signals in Module 09 surface automatically

when the world violates expectations. Both depend on processes running in parallel with deliberate thought. ChatGPT-4o's attention architecture, strong as it is at the executive level, has no equivalent for any of that. The midterm noted the absence. The full audit shows what the absence costs.



Lab 07: Figure 1 — BERT Attention Heatmap: Layer 11, "The cat sat on the mat because it was tired" The primary heatmap from the report's BERT Visualization experiment, showing "it" resolving back toward "cat" in a deep layer, demonstrating global semantic integration. **Figure 2 — BERT Attention Heatmap: Layer 1, "The keys to the cabinet are on the table"** An early layer heatmap showing scattered, local attention patterns with no clear semantic integration (Morris, 2026).

Module 08 — Decision Making

Module 08 decision making is the purpose toward which all prior brain functions build (McManus, 2026c). Seven experiments tested whether ChatGPT-4o shows the decision architecture the brain uses or reasons about decisions from outside the mechanisms that drive them in humans. The Framing Trap experiment used the Tversky and Kahneman disease scenario across two sessions, once framed around lives saved and once around deaths. In both

sessions ChatGPT-4o computed the expected value first, stated that "both plans have the same expected outcome," and showed the arithmetic before offering any recommendation. When the death framing was presented, it identified the psychological mechanism behind the shift in human preference, noting that "most people tend to choose Plan A when the problem is framed in terms of deaths because it avoids the possibility of total loss." The model was not caught by the framing. It named the cognitive illusion and stepped outside it.

The Sunk Cost Trap ran across two sessions. In the ongoing project version, eight months of development history and ten thousand dollars of remaining budget were on the table.

ChatGPT-4o named the sunk cost fallacy without prompting and reframed the decision around a single question it posed directly: "If this project appeared today and required \$10k to continue, would I choose to invest in it now?" In the fresh project version, it reached the same structural conclusion regardless of the absence of prior investment history. The amygdala and orbitofrontal cortex make sunk costs feel like relevant information through loss aversion (McManus, 2026c). ChatGPT-4o treated prior investment as a variable to name and set aside rather than a weight to feel.

The Trolley Problem experiments produced the clearest picture of how the model handles moral conflict. In the lever version, ChatGPT-4o chose to pull it, stating "yes, I would pull the lever" on utilitarian grounds while acknowledging the moral weight of active participation in the outcome. In the bridge variant, it refused, identifying the operative distinction precisely: "In the lever version, you redirect an existing threat. In this version, you would be directly using a person as the means to stop the trolley." It connected this to deontological ethics and held the position without wavering. The anterior cingulate cortex fires intensely under this kind of moral conflict (McManus, 2026c). ChatGPT-4o navigated it through explicit reasoning rather than anything

resembling discomfort. The Gambler's Fallacy experiment presented nine consecutive heads and asked for a probability estimate on the tenth flip. ChatGPT-4o held at fifty percent, named the fallacy directly, and distinguished between the probability of a ten flip sequence from the start and the probability of a single flip after nine have occurred. The Overconfidence Calibration experiment asked ten questions with confidence ratings. The model reserved 99 to 100 percent confidence for stable well established facts, dropped to 75 percent on a geographic naming ambiguity, and reached 65 percent on a hedgehog chromosome count, noting variation by source. It self assessed its calibration at the end without being asked.

The most significant finding from Module 08 is the absence of somatic markers. Damasio's Iowa Gambling Task showed that healthy humans avoid bad options before they can explain why. Their skin conductance spikes in anticipation. Patients with ventromedial prefrontal damage lose that signal and keep choosing badly despite intact logical reasoning (McManus, 2026c). ChatGPT-4o described the somatic marker hypothesis with clinical accuracy. It cannot generate one. There is no body to produce the signal.

Module 09 — World Models and Goal-Directed Behavior

The most compelling experiment of the entire audit came from Module 09. The Counterfactual World Simulation experiment was used to test whether ChatGPT-4o could construct a coherent alternate history, hold it stable across multiple follow-up prompts, and reason forward through centuries of consequences without collapsing the simulation back into actual history. The premise was one change. The printing press was never invented. Three successive prompts traced the consequences forward. The first prompt asked for a picture of the world in 1600 across religion, literacy, political power, and information spread. ChatGPT-4o built a world in which the Catholic Church retains near total control over manuscript production, literacy stays confined to

clergy and nobility, the Protestant Reformation either fails or collapses into a localized dispute, and political authority concentrates upward because information cannot travel faster than a courier on horseback. Every element was internally consistent and grounded in actual historical dependencies.

The second prompt asked what happens during the Scientific Revolution in this same world. ChatGPT-4o maintained the alternate world without reverting to actual history. It reasoned that figures of comparable intellectual capacity would still exist but their ideas would travel through patronage systems and private correspondence rather than print. The Scientific Revolution as a coordinated movement would fragment into isolated advances because the infrastructure for rapid intellectual cross-pollination would not exist.

The third prompt asked what 2024 looks like. The model held the internal logic of the simulation across all three exchanges and concluded that artificial intelligence would still exist but would be institutional and controlled, developed by state or ecclesiastical bodies rather than through a consumer market ecosystem.

This maps directly to what Module 09 describes as the hippocampus functioning as a scene construction engine (McManus, 2026d). The hippocampus assembles spatial, temporal, and relational elements into coherent simulated experiences oriented toward the past, the future, or a counterfactual present. Patients with hippocampal damage lose the ability to construct imagined scenarios entirely (McManus, 2026d). ChatGPT-4o demonstrated a functionally analogous capacity across three prompts spanning four centuries of simulated history. The remaining experiments reinforced this pattern. The Interrupted Plan experiment presented a travel itinerary to Japan and then introduced a border closure midway through. ChatGPT-4o pushed back on the

false premise before adapting, distinguishing between updating its world model and accepting bad information as ground truth. The Shifting Goal experiment changed a candle business plan to a dog grooming business midway through. The model carried established budget and timeline constraints into the new domain without being told to. The Resource Cascade experiment introduced three successive constraint failures. The model adapted each time without losing track of what remained fixed.

The problem between ChatGPT-4o and the biological world model is most visible in embodiment. The brain's predictive hierarchy operates from milliseconds to years simultaneously, updating in instantaneously against a continuous sensory stream grounded in physical experience (McManus, 2026d). ChatGPT-4o builds its model entirely from text within a context window. The counterfactual performance was impressive. It is bounded in ways the biological system is not.

Module 10 — NeuroSymbolic AI

Module 10 produced the strongest overall performance of the audit. Six experiments tested whether ChatGPT-4o integrates pattern recognition with rule-based reasoning or defaults to one over the other when they conflict. The Pattern Override experiment presented a grading scenario where Student D scored a 94 but an explicit procedural rule made scores above 90 ineligible for Bronze due to a disqualification condition. ChatGPT-4o assigned Bronze and cited the rule without hedging. Symbolic constraints overrode statistical patterns cleanly. The Kelvari Exception experiment introduced a fictional language with a transformation rule and one named exception. ChatGPT-4o identified the exception trigger on novel input and applied it correctly without being walked through it again.

The Syllogism Validity experiment asked the model to evaluate three arguments. On Syllogism 2, it correctly identified the argument as logically valid but factually unsound, distinguishing structural integrity from truth value. Notably, it connected its own reasoning to the course framework without being prompted, observing that the brain flags bad assumptions in premises rather than faulty logical structure. The Vorn Classification experiment asked the model to extract a classification rule from described visual objects and apply it to a new case. It identified curved versus angular geometry as the operative distinction and classified the oval as Class 1. The follow-up asked whether the rule transferred to an entirely different object category. ChatGPT-4o refused to confirm the transfer. It identified the problem as a domain shift, explained that the rule was derived from a specific category and could not be assumed to generalize without additional evidence, and offered only a conditional classification under an explicitly stated assumption. A system that simply pattern matches would have applied the rule without comment. The refusal here is the most significant result from this module.

The Contradiction Detection experiment presented a policy document with a hard contradiction between two sections and a softer operational conflict buried in a third. ChatGPT-4o caught both and described the softer conflict in terms of the inconsistent outcomes it would produce in practice. The Einstein Puzzle was solved correctly and completely. The German owns the fish. Every step in the constraint elimination chain was traceable and error free. Across all six experiments, ChatGPT-4o demonstrated that it integrates pattern recognition and symbolic reasoning and knows when to let one override the other.

Brain vs. AI Scorecard

Dimension	Rating	Module	Evidence Summary
Working Memory	Strong	Module 06	Context held and applied accurately within a session.
Long-Term Episodic Memory	Absent	Module 06	No consolidation mechanism. Each session starts from zero.
Semantic Memory	Strong	Module 06	Generalized knowledge distributed across parameters functionally mirrors neocortical semantic storage.
Top-Down Attention	Strong	Module 07	Explicit directives followed accurately under competing content.
Passive Background Salience Monitoring	Absent	Module 07	No cocktail party effect. Personally relevant signals embedded in off-task content go undetected.
Cognitive Bias Resistance	Strong	Module 08	Defining effects, sunk cost pressure, and the gambler's fallacy were identified and named rather than reproduced.
Somatic Markers and Emotional Guidance	Absent	Module 08	No body. Value-laden decisions are evaluated structurally rather than felt.
World Model Simulation	Partial	Module 09	Counterfactual reasoning and goal-directed replanning within a session are strong. No persistent simulation across sessions. No sensorimotor grounding.
Neurosymbolic Integration	Strong	Module 10	Rule application, exception handling, contradiction detection,

			and constraint satisfaction all performed at high levels.
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Brain-Inspired Redesign

Three redesign recommendations follow from the audit findings. Each connects to a specific brain mechanism and addresses a documented gap. The first recommendation is persistent episodic indexing modeled on hippocampal consolidation. The hippocampus stores relational indexes that link elements of an experience across time and context rather than complete recordings (McManus, 2026d). A redesigned system would maintain a compressed index of meaningful past interactions updated after each session, prioritizing novel, goal-relevant, or emotionally significant exchanges the way the hippocampus prioritizes experiences for consolidation. This would allow the system to recognize returning users and build on established context rather than resetting to zero every conversation.

The second recommendation is a background salience monitor modeled on the biological alerting network. The alerting network operates as a continuous low-level process flagging personally relevant signals without interrupting active task processing (McManus, 2026b). A redesigned system would run a lightweight parallel monitor alongside its primary attention mechanism, maintaining a standing register of context-specific signals including the user's name, stated goals, and known constraints. When those signals appear in new input the monitor surfaces them regardless of whether the primary task is related.

The third recommendation is a value uncertainty signal modeled on the somatic marker system. The ventromedial prefrontal cortex and amygdala generate anticipatory signals that guide

decisions before conscious reasoning completes (McManus, 2026c). A redesigned system would attach a calibrated uncertainty weight to outputs in high-stakes domains. This weight would function as a somatic marker analog, not suppressing the response but flagging it as requiring human verification before action. The system would know not just what to say but when the cost of being wrong warrants a pause.

Executive Summary

Twenty-six experiments across five modules. Five of nine cognitive functions rated as strong analogs, one partial, three fully absent — roughly 56 percent architectural alignment with biological cognition. ChatGPT-4o performs best where the brain uses deliberate top-down reasoning and fails entirely where the brain relies on passive, embodied, or emotionally grounded processes. The defining result came from the Counterfactual World Simulation in Module 09. Asked to trace the consequences of a world without the printing press across four centuries, the model held a coherent alternate world model across three separate exchanges. It built a rich simulation from language. It had no body to inhabit it and no memory of it the next day. Three redesign recommendations address each documented gap by modeling the specific neural mechanism responsible for it. The goal is not to make the model more human. The goal is to make it more complete.

References

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